Phase 2: Standalone ALU Validation

Objective

Build and validate the 8-bit arithmetic logic unit independently of the sequencer, verifying all computational functions and flag generation.

Component BOM

Qty	Component	Function
1	74F382N	8-bit ALU (2x 4-bit sections)
2	SN74LS273N	8-bit D-type registers (Register A and B)
1	DM74LS374N	Output latch for ALU result
1	74LS74	D flip-flops for flag capture (Zero/Carry)
8	LEDs	ALU result display
2	LEDs	Flag status display (Zero, Carry)
10	220-470Ω resistors	LED current limiting
16	DIP switches	Manual data input (8 for Reg A, 8 for Reg B)
4	DIP switches	ALU function select
16	10kΩ resistors	Pull-down for DIP switches
2	Pushbuttons	Manual clock (debounced)
1	Pushbutton	Reset
6	0.1μF capacitors	Power supply decoupling
1	4.7μF capacitor	Bulk power filtering
-	Breadboards, jumper wires	Construction
1	5V power supply	System power

ALU Function Table (74F382N)

S3	S2	S1	S0	Function	Description
0	0	0	0	CLEAR	F = 0000
0	0	0	1	B MINUS A MINUS 1	F = B - A - 1
0	0	1	0	A MINUS B MINUS 1	F = A - B - 1
0	0	1	1	ADD	F = A + B
0	1	0	0	A XOR B	F = A ⊕ B
0	1	0	1	B MINUS A	F = B - A
0	1	1	0	A MINUS B	F = A - B
0	1	1	1	A OR B	F = A + B
1	0	0	0	A AND B	F = A • B
1	0	0	1	A EQUALS B	F = A ⊙ B
1	0	1	0	B	F = B
1	0	1	1	A PLUS B PLUS 1	F = A + B + 1
1	1	0	0	A PLUS 1	F = A + 1
1	1	0	1	B PLUS 1	F = B + 1
1	1	1	0	A	F = A
1	1	1	1	SET	F = 1111

Circuit Configuration

Input Section

DIP Switch Bank 1 (8 switches) → Register A inputs (D0-D7)
DIP Switch Bank 2 (8 switches) → Register B inputs (D0-D7)
DIP Switch Bank 3 (4 switches) → ALU function select (S0-S3)
Pushbutton 1 → Register A clock enable
Pushbutton 2 → Register B clock enable
Pushbutton 3 → Output latch clock enable

ALU Connections

Register A outputs (Q0-Q7) → ALU A inputs
Register B outputs (Q0-Q7) → ALU B inputs
Function select switches → ALU S0-S3 inputs
ALU outputs (F0-F7) → Output latch inputs (D0-D7)
ALU Carry Out → Flag register input
ALU OVR/Zero → Flag register input

Output Section

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Output latch outputs (Q0-Q7) → 8x LEDs (result display)
Flag register outputs → 2x LEDs (status display)
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Test Procedures

1. Register Loading Test

Objective: Verify data loads correctly into registers

Procedure:

1. Set Register A DIP switches to 0b10101010
2. Pulse Register A clock
3. Set Register B DIP switches to 0b01010101
4. Pulse Register B clock
5. Verify register outputs with logic analyzer

Success Criteria: Logic analyzer shows stable register outputs matching DIP switch inputs

2. Basic Addition Test

Objective: Verify ADD function works correctly

Procedure:

1. Load Register A = 0x0F (15 decimal)
2. Load Register B = 0x01 (1 decimal)
3. Set function select to 0011 (ADD)
4. Pulse output latch clock
5. Verify LEDs show 0x10 (16 decimal)
6. Verify Carry flag = 0

Expected Results:

- Result LEDs: 0b00010000
- Carry LED: OFF
- Zero LED: OFF

3. Overflow Test

Objective: Verify carry flag generation

Procedure:

1. Load Register A = 0xFF (255 decimal)
2. Load Register B = 0x01 (1 decimal)
3. Set function select to 0011 (ADD)
4. Pulse output latch clock
5. Verify LEDs show 0x00 (0 decimal)
6. Verify Carry flag = 1, Zero flag = 1

Expected Results:

- Result LEDs: 0b00000000
- Carry LED: ON
- Zero LED: ON

4. Logical Operations Test

Objective: Verify AND, OR, XOR functions

Test Matrix:

A	B	AND	OR	XOR
0x0F	0xF0	0x00	0xFF	0xFF
0xAA	0x55	0x00	0xFF	0xFF
0xFF	0xFF	0xFF	0xFF	0x00

5. Subtraction Test

Objective: Verify subtraction and borrow handling

Procedure:

1. Load Register A = 0x10 (16 decimal)
2. Load Register B = 0x01 (1 decimal)
3. Set function select to 0110 (A MINUS B)
4. Verify result = 0x0F (15 decimal)

6. Complete Function Sweep

Objective: Test all 16 ALU functions systematically

Test Values: A = 0x0F, B = 0x03 **Procedure**: Step through all function codes 0000-1111, record results

Logic Analyzer Setup

Channel Assignment

Channels	Signal	Purpose
0-7	Register A outputs	Input verification
8-15	Register B outputs	Input verification
16-19	Function select	Operation mode
20-27	ALU outputs (pre-latch)	Real-time computation
28-35	Latched outputs	Final result
36	Carry flag	Status verification
37	Zero flag	Status verification
38	Clock signal	Timing analysis
39	Reset signal	System state

Timing Analysis Points

- **Setup Time**: Register input to clock edge
- **Hold Time**: Register input after clock edge
- **Propagation Delay**: Register output to ALU output
- **Flag Delay**: ALU output to flag update
- **Latch Delay**: ALU output to latched output

Success Criteria

- [ ] All 16 ALU functions produce mathematically correct results
- [ ] Carry flag activates correctly for overflow conditions
- [ ] Zero flag activates only when result = 0x00
- [ ] No timing races or glitches during operation
- [ ] Clean signal transitions on logic analyzer
- [ ] Flags update within 50ns of ALU output change
- [ ] No spurious flag toggles during function changes

Performance Validation

Expected Timing (74F series)

- **Register setup time**: 5ns typical
- **ALU propagation delay**: 11ns typical
- **Output latch setup time**: 4ns typical
- **Total input-to-output delay**: <25ns

Benchmark Tests

1. **Maximum Clock Frequency**: Determine highest reliable operation speed
2. **Function Change Glitches**: Verify clean transitions between operations
3. **Power Consumption**: Measure current draw at 5V
4. **Temperature Stability**: Verify operation across temperature range

Troubleshooting Guide

Issue: Incorrect Results

- Verify ALU function select wiring (S0-S3)
- Check register clock timing
- Confirm proper power supply voltage (4.75V-5.25V)

Issue: Unstable Flags

- Check flag register clock timing
- Verify ALU carry chain connections
- Add additional decoupling capacitors

Issue: Timing Problems

- Reduce clock frequency
- Check setup/hold time margins
- Verify signal integrity on breadboard

Issue: Random Behavior

- Check for floating inputs
- Verify all ground connections

- Add pull-down resistors to unused inputs

Integration Notes for Phase 3

- Document maximum reliable clock frequency
- Note any timing constraints for integration
- Identify optimal register loading sequence
- Verify flag timing margins for conditional operations
- Record power supply current requirements

Documentation Deliverables

- Complete test results matrix for all 16 functions
- Logic analyzer timing diagrams
- Performance benchmark data
- Photos of working circuit
- Schematic with actual component values